✅ 1. Import Libraries & Sample Dataset

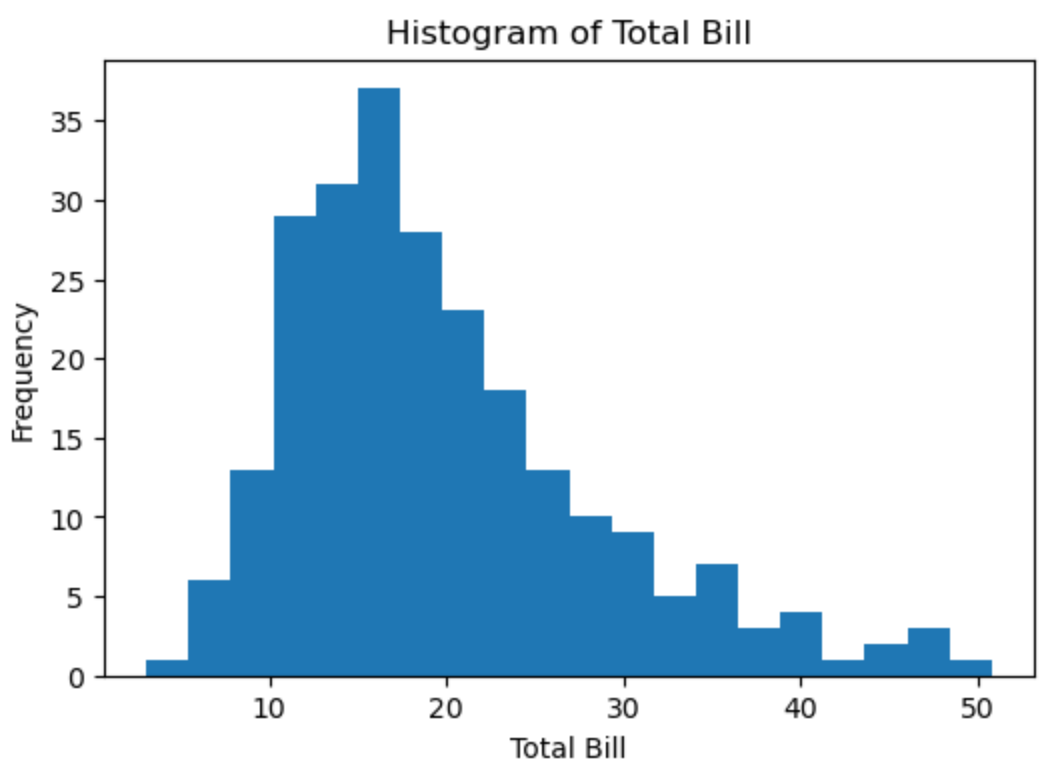
In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Sample dataset
df = sns.load_dataset("tips")   # built-in dataset
df.head()
```

Out[1]:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

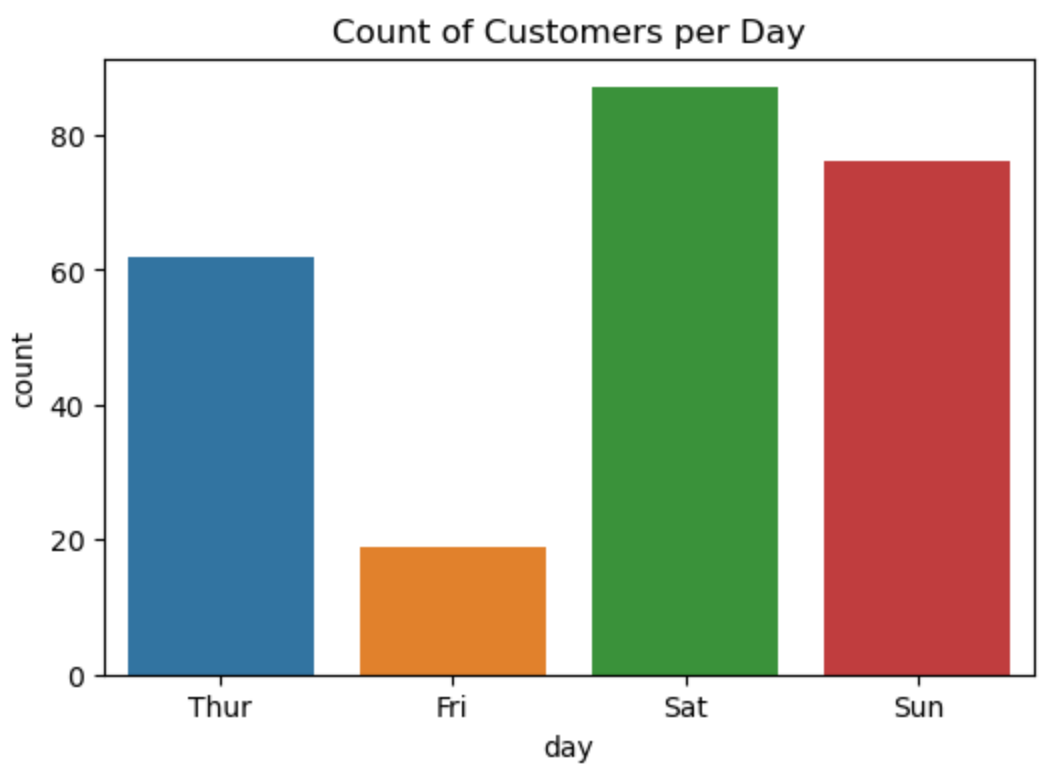
✅ 2. Histogram (Distribution Plot)

In [2]:
```
plt.figure(figsize=(6,4))
plt.hist(df['total_bill'], bins=20)
plt.title("Histogram of Total Bill")
plt.xlabel("Total Bill")
plt.ylabel("Frequency")
plt.show()
```



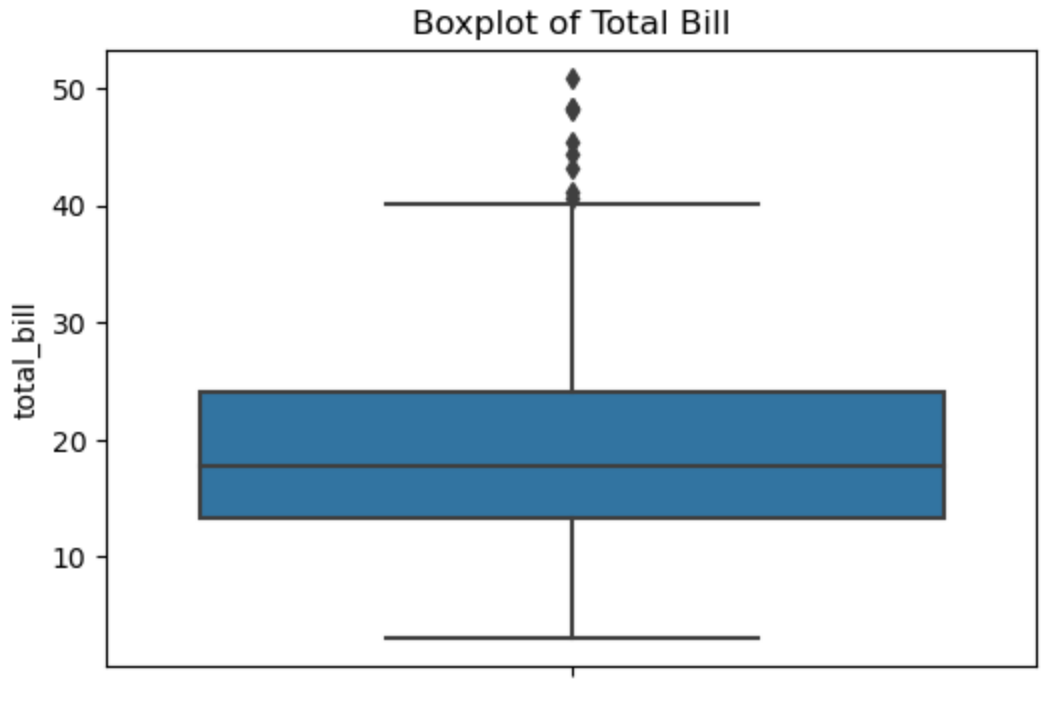
✅ 3. Bar Chart (Categorical Data)

In [3]:
```
plt.figure(figsize=(6,4))
sns.countplot(data=df, x="day")
plt.title("Count of Customers per Day")
plt.show()
```



✅ 4. Boxplot (Detect Outliers)

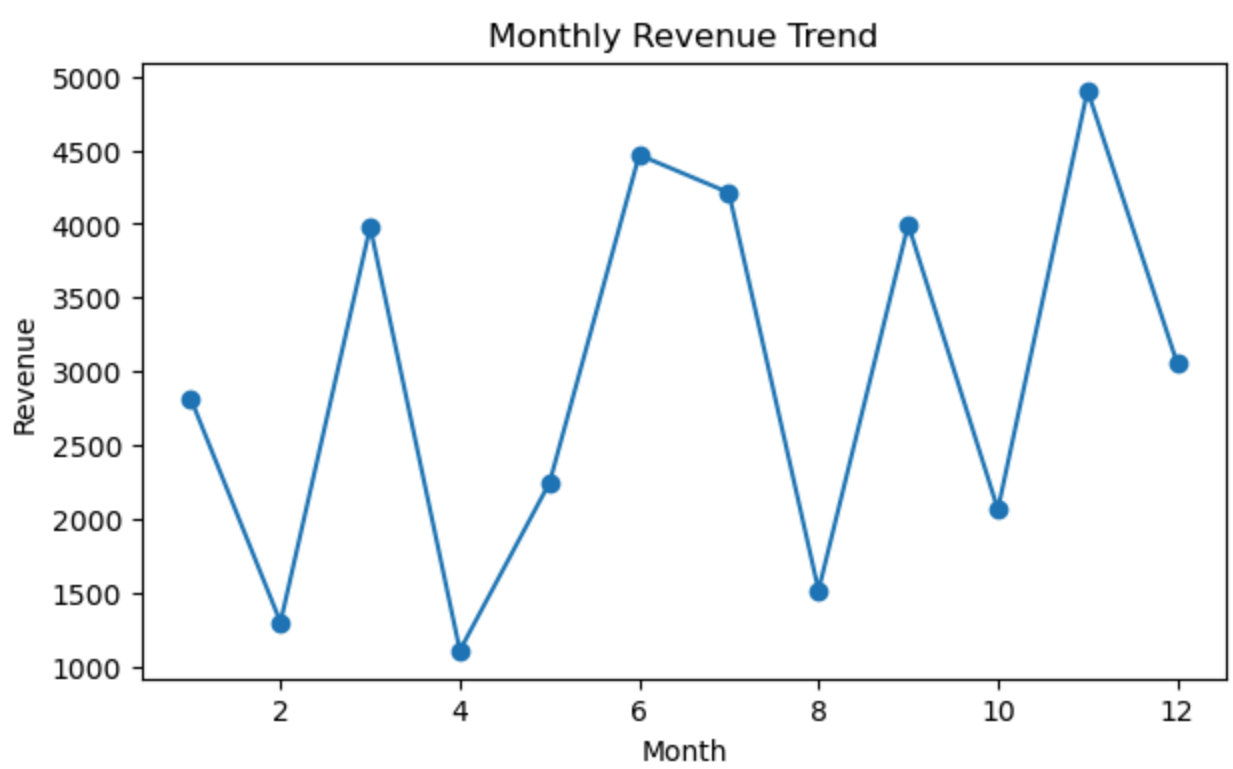
In [4]:
```
plt.figure(figsize=(6,4))
sns.boxplot(data=df, y="total_bill")
plt.title("Boxplot of Total Bill")
plt.show()
```



✅ 5. Line Chart (Trend Over Time)

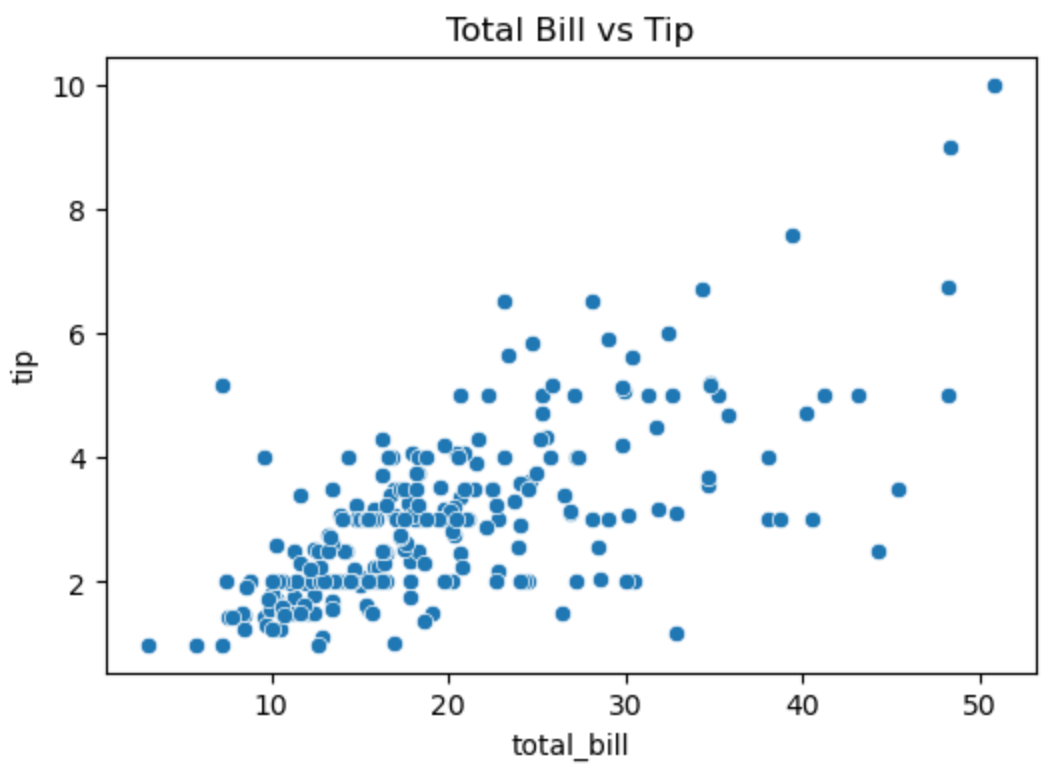
Example using simulated data:

In [5]:
```
sales = pd.DataFrame({
    "month": range(1,13),
    "revenue": np.random.randint(1000, 5000, 12)
})

plt.figure(figsize=(7,4))
plt.plot(sales['month'], sales['revenue'], marker='o')
plt.title("Monthly Revenue Trend")
plt.xlabel("Month")
plt.ylabel("Revenue")
plt.show()
```



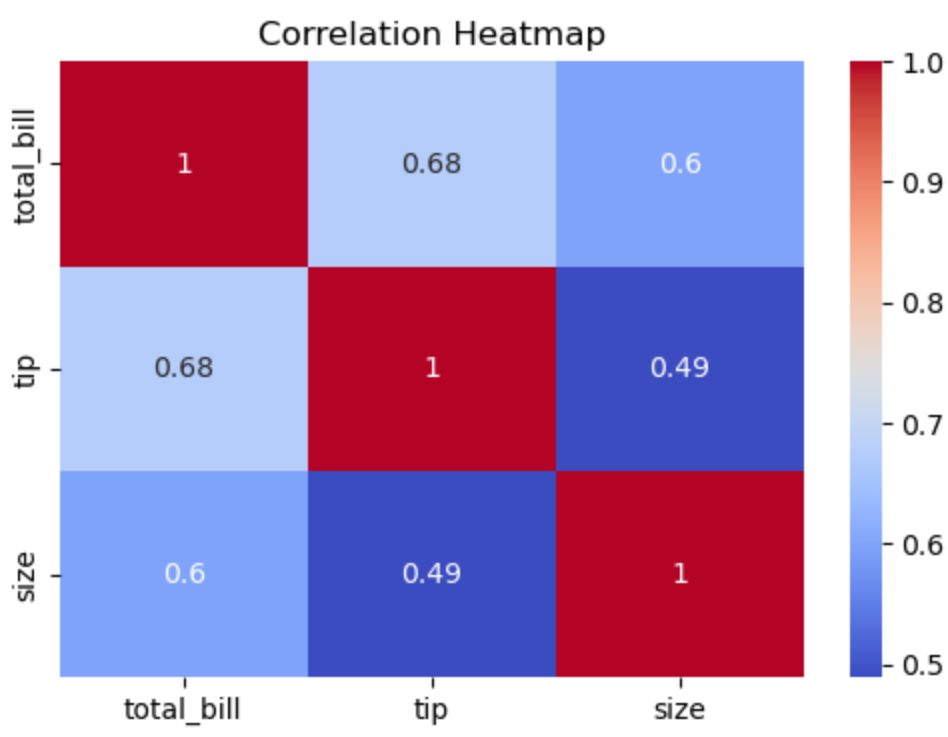
✅ 6. Scatter Plot (Relationship Between Variables)

In [6]:
```
plt.figure(figsize=(6,4))
sns.scatterplot(data=df, x="total_bill", y="tip")
plt.title("Total Bill vs Tip")
plt.show()
```



✅ 7. Correlation Heatmap

In [7]:
```
plt.figure(figsize=(6,4))
corr = df.corr(numeric_only=True)
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```

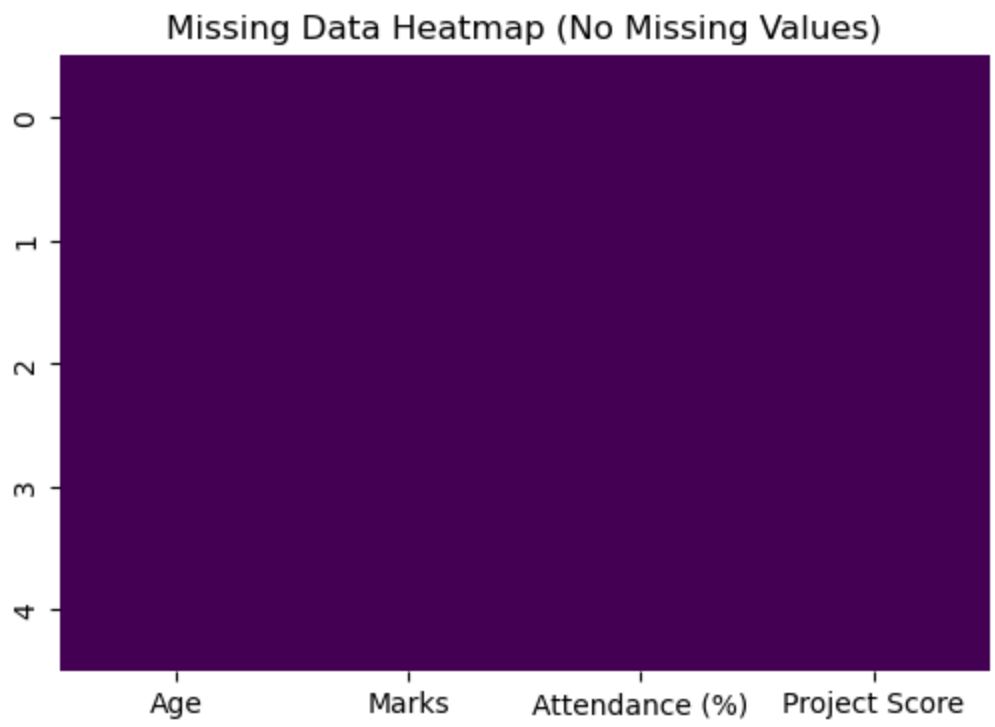


✅ 8. Missing Value Heatmap

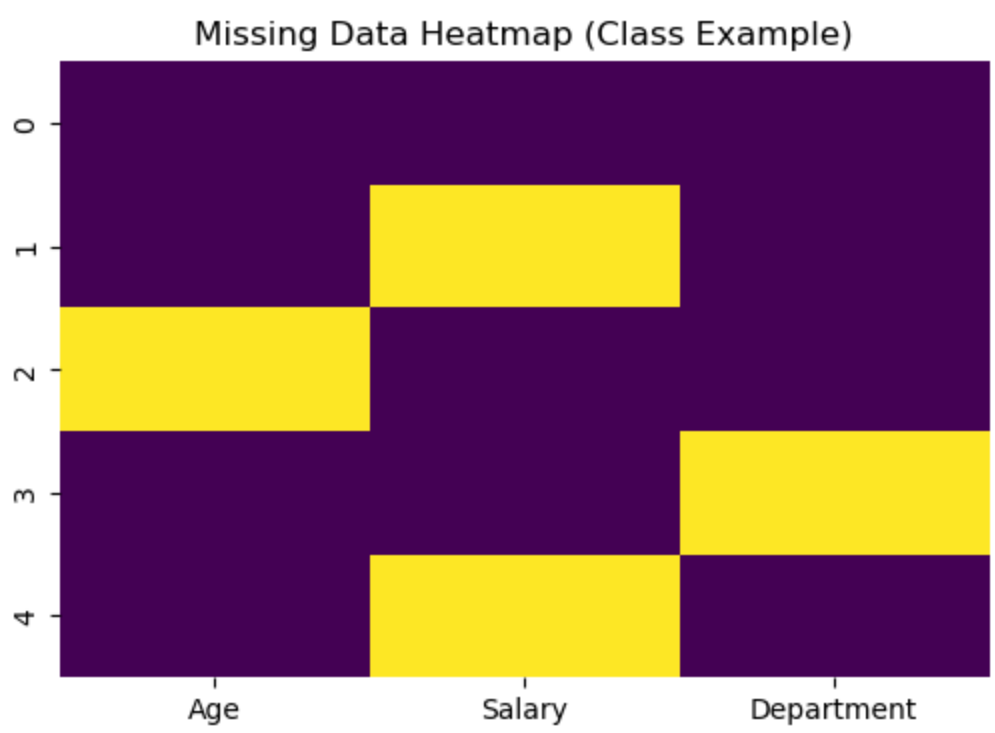
Bright color (yellow/green): missing values

Dark color: non-missing values

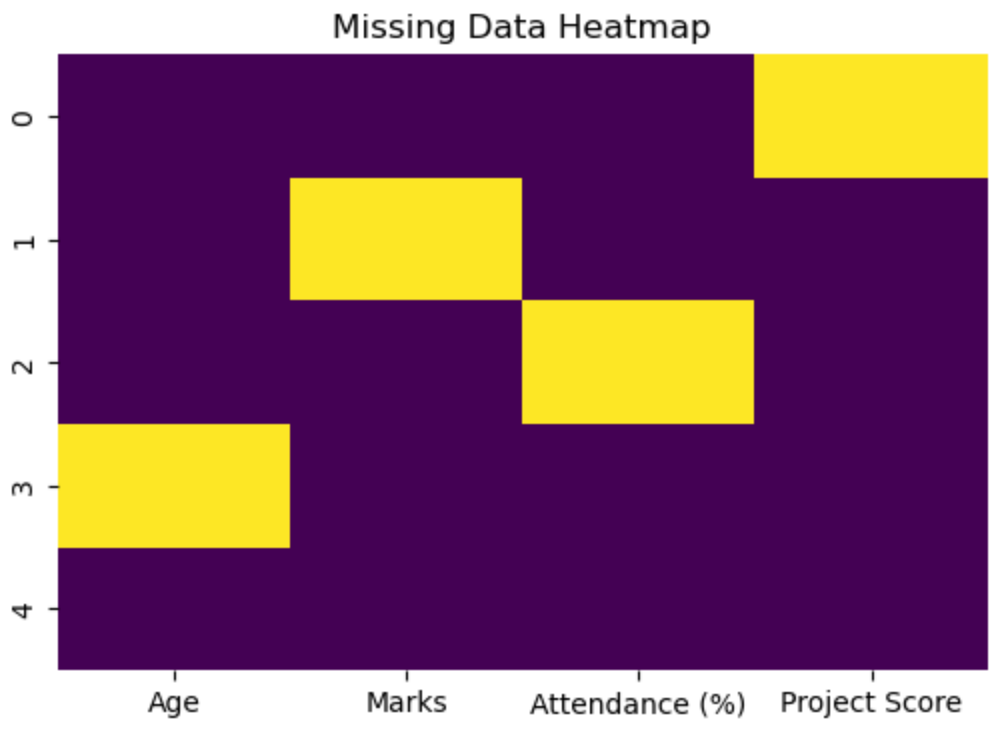
In [8]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Dataset with NO NaN values
data = {
    'Age': [18, 20, 19, 21, 22],
    'Marks': [85, 90, 78, 88, 92],
    'Attendance (%)': [92, 88, 95, 91, 93],
    'Project Score': [80, 85, 82, 88, 90]
}

df_clean = pd.DataFrame(data)

plt.figure(figsize=(6,4))
sns.heatmap(df_clean.isnull(), cbar=False, cmap="viridis")
plt.title("Missing Data Heatmap (No Missing Values)")
plt.show()
```



In [9]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# Sample dataframe with missing values
data = {
    'Age': [25, 30, np.nan, 35, 40],
    'Salary': [50000, np.nan, 60000, 55000, np.nan],
    'Department': ['HR', 'IT', 'Finance', np.nan, 'IT']
}

df2 = pd.DataFrame(data)

plt.figure(figsize=(6,4))
sns.heatmap(df2.isnull(), cbar=False, cmap="viridis")
plt.title("Missing Data Heatmap (Class Example)")
plt.show()
```



In [10]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# Simple dataset for teaching
data = {
    'Age': [18, 20, 19, np.nan, 22],
    'Marks': [85, np.nan, 78, 90, 88],
    'Attendance (%)': [92, 88, np.nan, 95, 91],
    'Project Score': [np.nan, 80, 85, 88, 90]
}

df = pd.DataFrame(data)

plt.figure(figsize=(6,4))
sns.heatmap(df.isnull(), cbar=False, cmap="viridis")
plt.title("Missing Data Heatmap")
plt.show()
```


Missing Data Heatmap
0
1
2
3
4
Age
Marks
Attendance (%)
Project Score

In [ ]: